(i)

$$\because \frac{d}{dx}\left(2x\sqrt{1+x^3}\right) = 2\sqrt{1+x^3} + 2x \cdot \frac{3x^2}{2\sqrt{1+x^3}}$$

$$\therefore \int \left(2\sqrt{1+x^3} + \frac{3x^3}{\sqrt{1+x^3}}\right) dx = 2x\sqrt{1+x^3} + C$$

(ii)

$$(x^2+2)\frac{\sin x}{x^3} = \frac{\sin x}{x} + \frac{2\sin x}{x^3}$$

$$\int \frac{2\sin x}{x^3}\,dx = -\frac{\sin x}{x^2} + \int \frac{\cos x}{x^2}\,dx$$

$$\int \frac{\sin x}{x}\,dx = -\frac{\cos x}{x} - \int \frac{\cos x}{x^2}\,dx$$

$$\int (x^2+2)\frac{\sin x}{x^3}\,dx = \int \frac{\sin x}{x}\,dx - \frac{\sin x}{x^2} - \frac{\cos x}{x} - \int \frac{\sin x}{x}\,dx$$

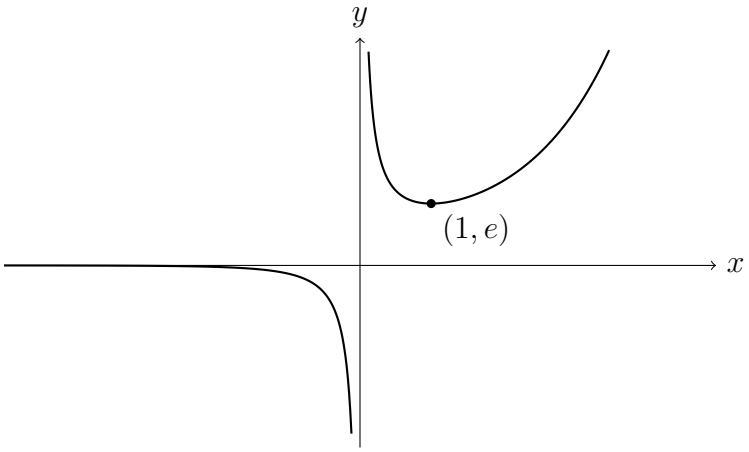
$$= -\frac{\sin x + x\cos x}{x^2} + C$$

(iii)(a)

$$\frac{dy}{dx} = \frac{e^x(x-1)}{x^2}$$

$$\frac{dy}{dx} = 0 \implies x = 1$$

x	$-\infty$	0^-	0^+	1	∞
y	0^-	$-\infty$	∞	e	∞



(iii)(b)

$$\int_a^{2a} \frac{e^x}{x}\,dx = \int_a^{2a} \frac{e^x}{x^2}\,dx$$

$$\because \frac{d}{dx}\left(\frac{e^x}{x}\right) = \frac{e^x}{x} - \frac{e^x}{x^2}$$

$$\left[\frac{e^x}{x}\right]_a^{2a} = 0$$

$$\frac{e^{2a}}{2a} - \frac{e^a}{a} = 0$$

$$a = \ln 2$$

(iii)(c) Assume distinct integers m, n exist and $n < m$.

$$\left[\frac{e^x}{x}\right]_m^n = 0 \implies \frac{e^n}{n} = \frac{e^m}{m}$$

From the graph in **(iii)(a)**, $\dfrac{e^n}{n}$ must lie in $(0, 1)$.

$\therefore$ Contradiction, no distinct integers m, n exist

This solution was submitted by **Jon**

Published and refined by **Jon**

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